Form D

# Core Focus on Proportions & Probability

## Block 2 Review ~ Proportions and Similarity

**ANSWER KEY**

Name _____ Period _____ Date _____

## Part I – Selected Response

**1.** Which ratios form a proportion with $\frac{12}{20}$? Circle all that apply.

A. $\frac{15}{20}$ B. $\frac{9}{15}$ C. $\frac{20}{12}$

D. $\frac{3}{5}$ E. $\frac{21}{35}$ F. $\frac{16}{24}$

B, D, E (cross products: $12 \cdot 15 = 20 \cdot 9 = 180$; $12 \cdot 5 = 20 \cdot 3 = 60$; $12 \cdot 35 = 20 \cdot 21 = 420$. A: $240 \neq 300$, C: $144 \neq 400$, F: $288 \neq 320$)

**2.** A 6-pack of yogurt costs \$4.50. A 10-pack of the same yogurt costs \$6.90. How much less per cup does the 10-pack cost?

A. \$0.06

B. \$0.69

C. \$0.75

D. \$2.40

A (\$4.50 ÷ 6 = \$0.75 per cup; \$6.90 ÷ 10 = \$0.69 per cup; \$0.75 − \$0.69 = \$0.06)

**3.** Ana walked 3 miles in 1 hour 15 minutes. At this pace, how long would it take her to walk 7 miles?

A. 2 hours 20 minutes

B. 2 hours 41 minutes

C. 3 hours 5 minutes

D. 2 hours 55 minutes

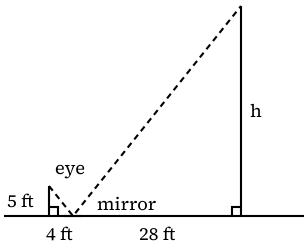
D (1 h 15 min = 75 min; $3/75 = 7/x \rightarrow 3x = 525 \rightarrow x = 175$ min = 2 h 55 min)

**4.** **For numbers 4a – 4e, circle YES or NO to tell whether the shapes given are always similar.**

| | | |
|---|---|---|
| 4a. two rectangles | YES | NO |
| 4b. two equilateral triangles | YES | NO |
| 4c. two right triangles | YES | NO |
| 4d. two circles | YES | NO |
| 4e. two squares | YES | NO |

4a NO (2 × 3 and 2 × 5 are not proportional) · 4b YES (all angles 60°, sides always in one ratio) · 4c NO (the other two angles can differ) · 4d YES · 4e YES (all angles 90°, all sides equal)

**5.** Maya used a mirror to find the height of a flagpole. Her eye height is 5 feet and she stands 4 feet from the mirror. The mirror is 28 feet from the base of the flagpole. How tall is the flagpole?




A. 22.4 feet

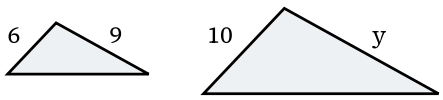
B. 29 feet

C. 35 feet

D. 140 feet

C (h/5 = 28/4 → 4h = 140 → h = 35 ft)

**6.** The triangles below are similar. Which proportions could be used to solve for $y$? Circle all that apply.




A. $\frac{6}{10} = \frac{9}{y}$ B. $\frac{6}{9} = \frac{y}{10}$

C. $\frac{10}{6} = \frac{y}{9}$ D. $\frac{9}{y} = \frac{10}{6}$

E. $\frac{y}{9} = \frac{5}{3}$ F. $\frac{6}{y} = \frac{10}{9}$

A, C, E (each gives $6y = 90$ or $3y = 45$, so $y = 15$; E uses the scale factor $10 : 6 = 5 : 3$. B gives $y = 6\frac{2}{3}$, D and F give $y = 5.4$)

**7. Two similar hexagons have perimeters of 18 cm and 30 cm. For numbers 7a – 7d, determine whether each statement about the hexagons is true or false.**

7a. Their scale factor is 3 : 5. **TRUE FALSE**

7b. The ratio of their areas is 9 : 25. **TRUE FALSE**

7c. Two corresponding sides could be 6 cm and 12 cm. **TRUE FALSE**

7d. If the smaller hexagon has an area of 18 $cm^2$, the larger has an area of 30 $cm^2$. **TRUE FALSE**

**7a TRUE (18 : 30 = 3 : 5) · 7b TRUE ($3^2 : 5^2$) · 7c FALSE (6 : 12 = 1 : 2) · 7d FALSE (9/25 = 18/x → x = 50 $cm^2$)**

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## Block 2 Review ~ Proportions and Similarity

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Name _____ Period _____ Date _____

## Part II – Constructed Response

**1.** Solve the proportion. $\frac{2.4}{x} = \frac{6}{15}$

x = 6 (6x = 2.4 · 15 = 36; check 2.4/6 = 0.4 = 6/15)

**2.** Jonah says $x = 9$ is the solution to $\frac{4}{10} = \frac{x}{25}$. Use cross products to decide whether he is correct. If he is not, find the correct value.

Not correct: 4 · 25 = 100 but 10 · 9 = 90, and 100 ≠ 90. Correct: 10x = 100 → x = 10.

**3.** In the school band, the ratio of brass players to woodwind players is 4 : 7. There are 28 woodwind players. How many brass players are in the band? Show all work necessary to justify your answer.

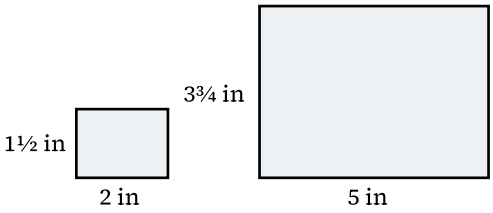
16 brass players (4/7 = b/28 → 7b = 112 → b = 16)

**4.** Store A sells 12 ounces of trail mix for $4.20. Store B sells 20 ounces of the same trail mix for $6.60. Which store has the better buy? How much would 30 ounces cost at the better price? Use mathematics to justify your answer.

Store B. A: $4.20 ÷ 12 = $0.35 per ounce. B: $6.60 ÷ 20 = $0.33 per ounce, which is less. 30 oz at Store B: 30 × $0.33 = $9.90.

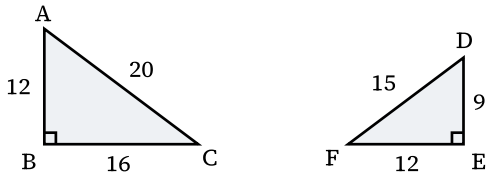
***Determine the scale factor for each pair of similar figures.***

**5.** Smaller to larger:



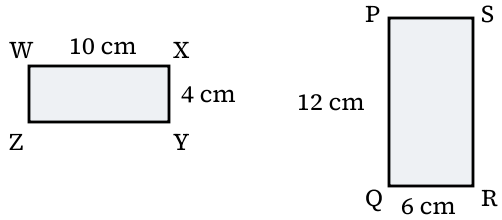

2 : 5 (2 : 5 and 1½ : 3¾ = 6 : 15 = 2 : 5)

**6.** From △ABC to △DEF:




4 : 3 (AB : DE = 12 : 9, BC : EF = 16 : 12, AC : DF = 20 : 15, all 4 : 3; DEF to ABC would be 3 : 4)

**7.** Both figures below are rectangles. Use the order of the letters to find the corresponding sides and angles.




$\overline{WX}$ corresponds to _____ ∠W corresponds to ∠_____

$\overline{XY}$ corresponds to _____ ∠X corresponds to ∠_____

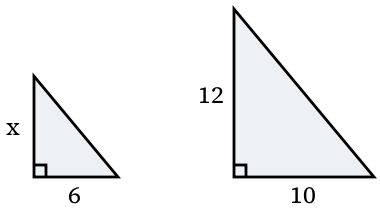
$\overline{YZ}$ corresponds to _____ ∠Y corresponds to ∠_____

Are rectangle WXYZ and rectangle PQRS congruent, similar, or neither? Explain your reasoning.

WX ↔ PQ, XY ↔ QR, YZ ↔ RS; ∠W ↔ ∠P, ∠X ↔ ∠Q, ∠Y ↔ ∠R. Neither: all the angles are 90°, but the sides are not proportional (10/12 = 5/6 but 4/6 = 2/3), so they are not similar, and not congruent.

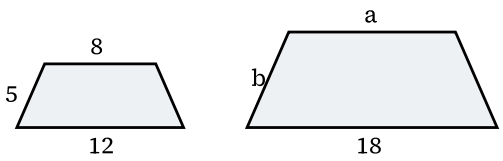
***Each pair of shapes below are similar. Solve for each variable. Show all work necessary to justify your answer.***

**8.**



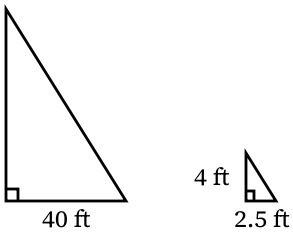

x = 7.2 (6/10 = x/12 → 10x = 72 → x = 7.2)

**9.**




Scale factor 12 : 18 = 2 : 3. a: 12/18 = 8/a → 12a = 144 → a = 12. b: 12/18 = 5/b → 12b = 90 → b = 7.5.

**10.** A 4-foot fence post casts a shadow 2.5 feet long. At the same time, a cell tower casts a shadow 40 feet long.



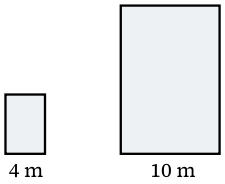

a. Find the height of the cell tower. Use mathematics to justify your answer.

64 feet (h/4 = 40/2.5 → 2.5h = 160 → h = 64)

b. Liam measured the tower's shadow at noon and the post's shadow at 4 p.m. Explain why he cannot use these two shadows in a proportion.

The sun's angle changes during the day, so shadows measured at different times do not form similar triangles. Both shadows must be measured at the same time.

**11.** Use the similar rectangles below.




a. Find their scale factor (small to large).

2 : 5 (4 : 10)

b. Find the ratio of their perimeters.

2 : 5

c. Find the ratio of their areas.

4 : 25 ($2^2 : 5^2$)

d. The larger rectangle has an area of 150 $m^2$. What is the area of the smaller rectangle? Show all work necessary to justify your answer.

24 $m^2$ (4/25 = x/150 → 25x = 600 → x = 24; check 4 × 6 = 24 and 10 × 15 = 150)

**12.** Two similar triangles have perimeters of 9 feet and 15 feet. The larger triangle has an area of 50 $ft^2$. Omar used this information to find the area of the smaller triangle. Unfortunately, he made an error in his work. Explain the error and then find the correct area of the smaller triangle.

Scale factor is 9 : 15, which is 3 : 5

Ratio of their areas is $3^2 : 5^2$, which is 9 : 25

Let $x$ = the area of the smaller triangle

$\frac{9}{25} = \frac{50}{x}$

$x \approx 138.9$

The area of the smaller triangle is about 138.9 $ft^2$.

Omar put the larger area (50) with the smaller triangle's part of the ratio (9). The areas must be in the same order as the ratio: 9/25 = x/50 → 25x = 450 → x = 18. The smaller area is 18 $ft^2$. (His answer is bigger than 50, which cannot be right for the smaller triangle.)

**13.** On a map, two towns that are 84 miles apart are 3½ inches apart.

a. Find the map's scale as 1 inch : ____ miles.

1 inch : 24 miles (3.5/84 = 1/x → 3.5x = 84 → x = 24)

b. Two lakes are 2¼ inches apart on the same map. How many miles apart are the lakes? Show all work necessary to justify your answer.

54 miles (1/24 = 2.25/x → x = 54)

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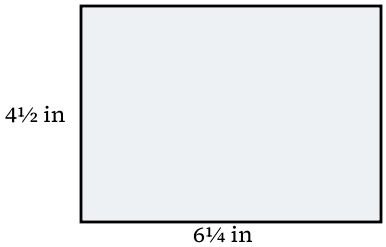
**ANSWER KEY**

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## Part III – Problem Solving

Ms. Ortega's blueprint for a new game room has a scale of 1 inch : 4 feet. On the blueprint, the room is 6¼ inches long and 4½ inches wide. New flooring costs $3.20 per square foot. How much will it cost to cover the whole floor?

Dev tried this problem. He found the blueprint area, 6¼ × 4½ = 28⅛ square inches, and multiplied by 4 to get a floor area of 112½ square feet. Explain Dev's mistake, then find the correct cost. Finally, a pool table in the room will be 9 feet long. How long should it be drawn on the blueprint? Show all work.




**Length: 1/4 = 6.25/L → L = 25 ft. Width: 1/4 = 4.5/W → W = 18 ft. Area = 25 × 18 = 450 ft². Cost = 450 × $3.20 = $1,440. Dev's mistake: the scale works on lengths, not areas. Each length is 4 times larger, so the area is 4² = 16 times larger (28⅛ × 16 = 450), not 4 times. Scale each length first, then multiply. Pool table: 1/4 = x/9 → 4x = 9 → x = 2¼ in.**